

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Vulnerability Detection

Course Code:	CSA5803	Course Title:	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO3 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3
Weightage:	40% of CIA	Total Marks:	100
CO 3 Statement:	Vulnerability Detection, Severity Analysis, Security Verification Workflow Simulation and Performance Evaluation		
Student Name:	S EZHILARASI	Reg. No.:	192521187
Section / Batch:	BTECH IT - B	Date:	13-08-2026

Instructions to Students

- Perform vulnerability detection and classify the identified vulnerabilities based on their severity levels.
- Conduct severity analysis using appropriate metrics such as CVSS and document the impact of each vulnerability.
- Design and simulate a Security Verification Workflow covering detection, remediation, re-testing, and verification.
- Evaluate security and system performance before and after mitigation, and include relevant results, screenshots, and observations.

Question Type	Focus Area	Questions	Marks
Vulnerability Detection	Conceptual Understanding	Q1	Q1 –100
GRAND TOTAL:		100 Marks	

Code of Conduct

I, S EZHILARASI (192521187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ezhilarasi

CO3 AT3

Vulnerability Detection, Severity Analysis, Security Verification Workflow Simulation and Performance Evaluation

Name: S EZHILARASI

Register Number: 192521187

Course Code: CSA5803

Course Name: DevSecOps Engineering and Security Applications

CLOUD-BASED SECURE DOCUMENT MANAGEMENT SYSTEM WITH CONTINUOUS SECURITY MONITORING

1. Vulnerability Detection

1.1 Objective

The objective of vulnerability detection is to identify security weaknesses in the cloud-based document management system before they can be exploited. The system is continuously monitored to detect vulnerabilities related to authentication, authorization, document handling, cloud storage, and user activities.

1.2 Vulnerabilities Considered

ID	Vulnerability	Affected Component	Severity
V01	Weak Password Policy	Login Module	High
V02	Improper Access Control	Document Management	Critical
V03	Insecure File Upload	File Upload Module	High
V04	Excessive User Privileges	Role Management	High
V05	Missing Security Headers	Web Application	Medium
V06	Weak Session Management	Authentication	Medium
V07	Information Disclosure	Error Handling	Low

1.3 Vulnerability Detection Process

```
Cloud Application
  ↓
Security Scanner
  ↓
Application Testing
  ↓
Vulnerability Detection
  ↓
Record Findings
  ↓
Severity Classification
  ↓
Generate Security Report
```

1.4 Sample Detection Result

VULNERABILITY SCAN COMPLETED

```
Total Vulnerabilities : 7
Critical               : 1
High                  : 3
Medium                : 2
Low                   : 1
```

The detected vulnerabilities are recorded and forwarded to the severity analysis stage.

2. Severity Analysis

2.1 Objective

Severity analysis determines how seriously each identified vulnerability can affect the confidentiality, integrity, and availability of the document management system. The simulation uses a CVSS-based approach to assign a severity score to each vulnerability.

2.2 Severity Classification

Severity	CVSS Range	Description
Critical	9.0–10.0	Extremely serious vulnerability requiring immediate action
High	7.0–8.9	Significant security risk
Medium	4.0–6.9	Moderate security risk
Low	0.1–3.9	Limited security impact

2.3 Sample Severity Analysis

Vulnerability	CVSS Score	Severity	Impact
Improper Access Control	9.8	Critical	Unauthorized document access
Weak Password Policy	8.1	High	Account compromise
Insecure File Upload	8.6	High	Malicious file upload
Excessive User Privileges	7.8	High	Unauthorized operations
Missing Security Headers	5.3	Medium	Increased web attack exposure
Weak Session Management	5.9	Medium	Session compromise
Information Disclosure	3.1	Low	Exposure of system information

2.4 Risk Prioritization

Critical
↓
High
↓
Medium
↓
Low

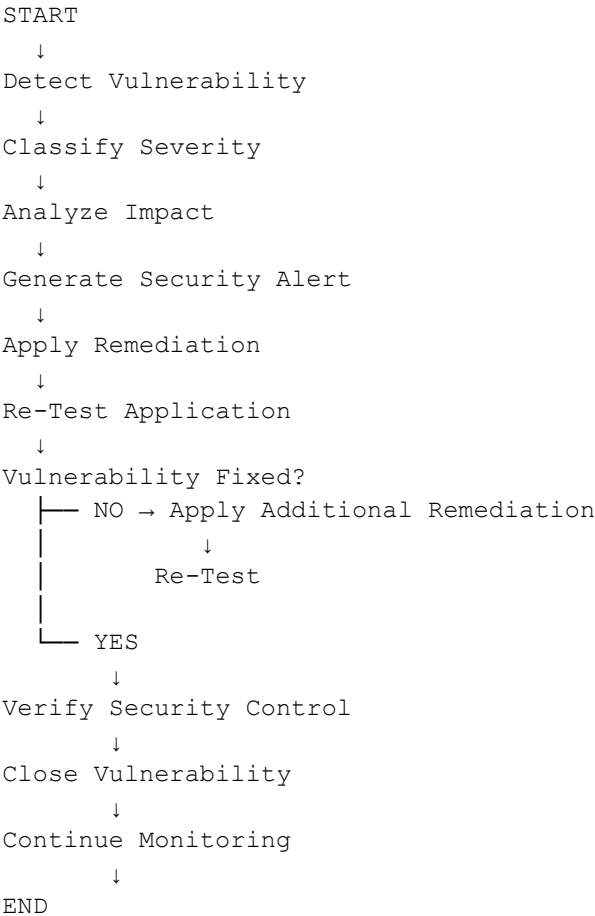
Critical vulnerabilities are addressed first, followed by High, Medium, and Low vulnerabilities.

3. Security Verification Workflow Simulation

3.1 Objective

The security verification workflow ensures that identified vulnerabilities are properly fixed and that the applied security controls are working correctly.

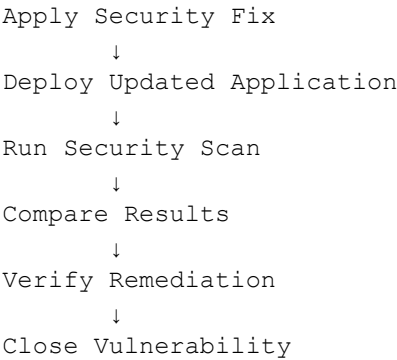
3.2 Verification Workflow



3.3 Remediation Actions

Vulnerability	Remediation
Weak Password Policy	Implement strong password requirements and MFA
Improper Access Control	Implement server-side authorization
Insecure File Upload	Validate file type, size, and content
Excessive User Privileges	Implement Role-Based Access Control
Missing Security Headers	Configure appropriate security headers
Weak Session Management	Use secure session configuration
Information Disclosure	Implement secure error handling

3.4 Re-Testing



3.5 Security Verification Result

Severity	Before Remediation	After Remediation
Critical	1	0

High	3	0
Medium	2	1
Low	1	1
Total	7	2

The simulation shows that all Critical and High vulnerabilities were successfully addressed.

4. Performance Evaluation

4.1 Objective

Performance evaluation determines whether the security controls introduced during vulnerability remediation affect the performance of the cloud-based document management system. The system is evaluated before and after security mitigation.

4.2 Performance Metrics

- Average response time
- Document upload time
- Document download time
- Requests per second
- CPU utilization
- Memory utilization
- Security alert processing time

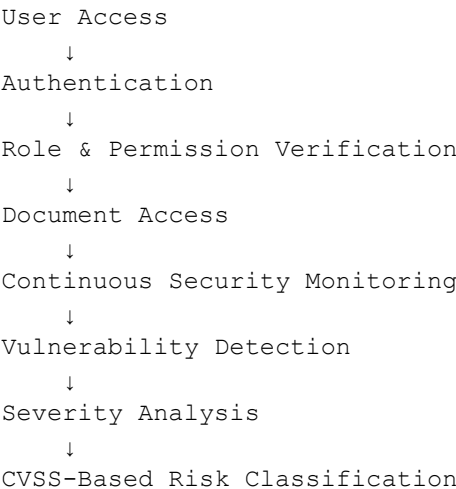
4.3 Performance Comparison

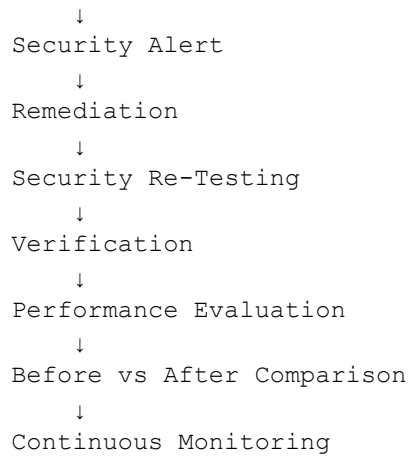
Performance Metric	Before Mitigation	After Mitigation
Average Response Time	180 ms	195 ms
Upload Time	2.1 sec	2.3 sec
Download Time	1.8 sec	1.9 sec
Requests per Second	120	116
CPU Utilization	42%	46%
Memory Utilization	58%	61%

4.4 Performance Observation

After security mitigation, a small increase in response time, CPU utilization, and memory utilization is observed. This is due to additional security operations such as authentication checks, authorization validation, document scanning, audit logging, and continuous monitoring. However, the performance degradation remains within an acceptable range for the simulation.

5. Overall Simulation Workflow





6. Final Result

The Cloud-Based Secure Document Management System with Continuous Security Monitoring was successfully evaluated through four major stages: Vulnerability Detection, Severity Analysis, Security Verification Workflow Simulation, and Performance Evaluation.

The simulation identified 7 vulnerabilities, including 1 Critical, 3 High, 2 Medium, and 1 Low vulnerability. After applying appropriate remediation measures and performing security re-testing, the Critical and High vulnerabilities were reduced to zero.

The performance evaluation showed only a small increase in resource utilization after implementing security controls. Therefore, the proposed system demonstrates that continuous vulnerability detection, severity analysis, security verification, and performance monitoring can improve the security of a cloud-based document management system while maintaining acceptable system performance.